

AWS VPC, PUBLIC SUBNET & EC2 INSTANCE SETUP

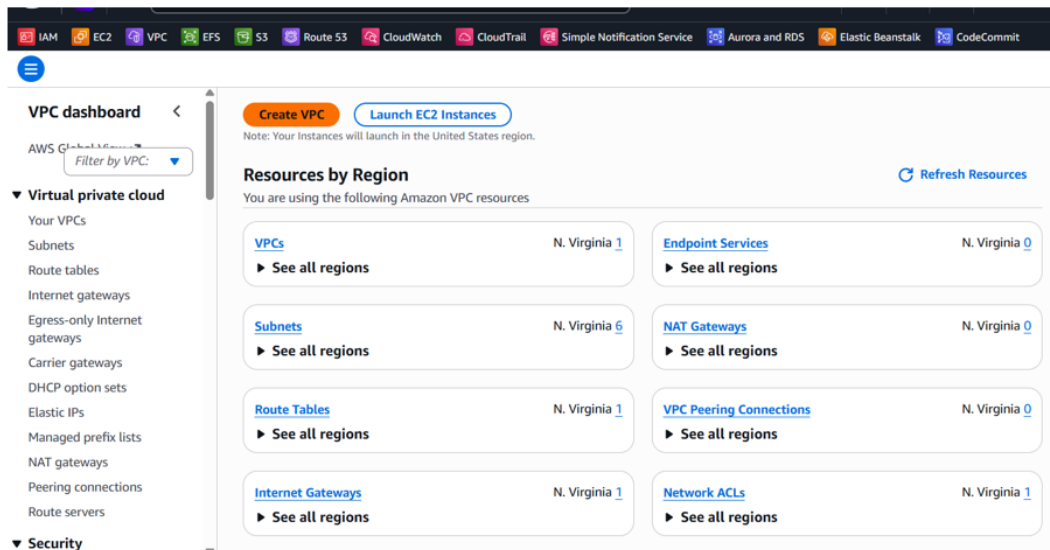
THIS PROJECT DEMONSTRATES THE BASIC SETUP OF AWS NETWORKING BY CREATING CUSTOM VPC ,A PUBLIC SUBNET,AN INTERNET GATEWAY AND A ROUTE TABLE

STEPS TO CREATE....

STEP:1 CREATE A VPC

- Open AWS Console.
- Go to **VPC Dashboard**.
- Click **Create VPC**.
- Enter VPC name.
- Enter IPv4 CIDR block (Example: (192.168.0.0/16)
- Click **Create VPC**.

➤ CLICK ON CREATE VPC



Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional [Info](#)
Creates a tag with a key of 'Name' and a value that you specify.

first-vpc09

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
192.168.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block

VPC dashboard [Subnets](#)

Subnets (6) [Info](#)

Last updated 4 minutes ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

	Default subnet	Auto-assign public IPv4 add...	Auto-assign customer-own...	Customer-owned IPv4 pool	Auto-assign IPv6 address
	Yes	Yes	No	-	No
	Yes	Yes	No	-	No
	Yes	Yes	No	-	No

Select a subnet

STEP :2 CREATE A PUBLIC SUBNET

- Go to **Subnets**.
- Click **Create Subnet**.
- Select the VPC.
- Enter subnet name.
- Choose an Availability Zone.
- Enter CIDR block (Example: 192.168.1.0/24).
- Click **Create Subnet**

[VPC](#) > [Subnets](#) > Create subnet

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-01db72eec4dcb2558 (my--vpc)

Associated VPC CIDRs

IPv4 CIDRs

192.168.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

my--subnet01

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az1 (us-east-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/16

IPv4 subnet CIDR block

192.168.1.0/24

256 IPs

VPC dashboard <

AWS Global Filter by VPC: ▾

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Subnets (1) [Info](#)

Last updated less than a minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

Subnet ID: subnet-0760a0ced36bab32b X Clear filters

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	my--subnet01	subnet-0760a0ced36bab32b	Available	vpc-01db72eec4dcb2558 my--...	Off	192.168.1.0/24

Select a subnet

STEP:3 Enable Auto-Assign Public IP

- Select the subnet.
- Choose **Actions** → **Edit Subnet Settings**.
- Enable **Auto-assign Public IPv4 Address**.
- Save changes.

▼ Network settings

Info

VPC - required

Info

vpc-01db72eec4dcb2558 (my--vpc)

192.168.0.0/16

▼

↻

Subnet

Info

subnet-0760a0ced36bab32b

my--subnet01

▼

↻

Create new subnet

VPC: vpc-01db72eec4dcb2558

Owner: 615167489280

Availability Zone: us-east-1a (use1-az1)

Zone type: Availability Zone

IP addresses available: 251

CIDR: 192.168.1.0/24)

Auto-assign public IP

Info

Enable

▼

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

STEP:4 : Create an Internet Gateway

- Go to **Internet Gateways**.
- Click **Create Internet Gateway**.
- Enter a name.
- Create it.
- Attach it to your VPC.

Create internet gateway

Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

internet-gateway 01

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name

X

Q internet-gateway 01

X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

Step 5: Create a Route Table

- Go to **Route Tables**.
- Create a new route table.
- Associate it with your VPC.
- Add a route:
 - **Destination:** 0.0.0.0/0
 - **Target:** Internet Gateway

- Associate the route table with the public subnet.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

route-table

VPC

The VPC to use for this route table.

vpc-01db72eec4dcb2558 (my-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q route-table

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Q local			
	Internet Gateway	-	No	CreateRoute
	Q igw-011c1e0102dd9343d			

Add route

Cancel

Preview

Save changes

STEP:6 CONNET TO THE EC2 INSTANCE

- SELECT THE EC2
- CLICK ON CONNECT
- CHOOSE EC2 INSTANCE CONNECT
- CLICL ON CONNECT

EC2 Instance Connect

SSM Session Manager

SSH client

EC2 serial console

Instance ID

i-090f609b03f4c8602 (EC2-instance-vpc-demo)

Connection type



Connect using a Public IP

Connect using a public IPv4 or IPv6 address



Public IPv4 address

44.201.4.126



IPv6 address

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2



ec2-user



```

      #
    ~\  #####      Amazon Linux 2023
    ~\  #####\
    ~\  #####|
    ~\  \#/      https://aws.amazon.com/linux/amazon-linux-2023
    ~\  V~'  '--->
    ~\  .-.-.-
    ~\  /_/_/_/
    ~\  /m/'
ec2-user@ip-192-168-1-64 ~]$
```

VERIFY CONNECTION RUN THE FOLLOWING COMMAND

- PWD
- WHOAMI
- HOSTNAME
- DATE